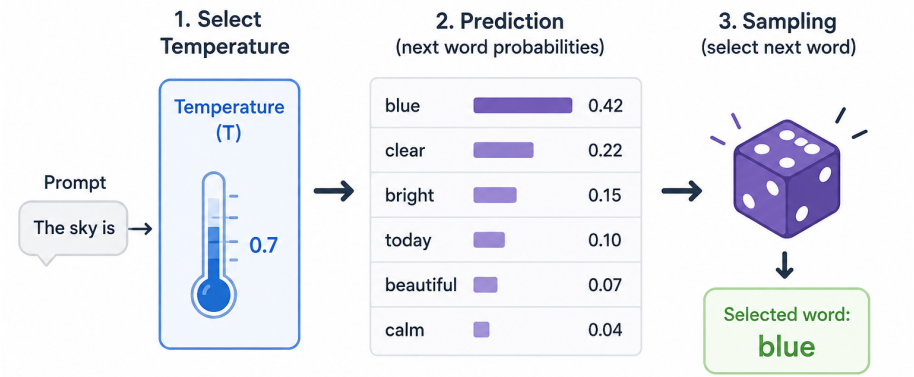


CORE CONCEPTS

Temperature & Sampling in LLMs

The knobs that control how your LLM thinks and why most people set them wrong.

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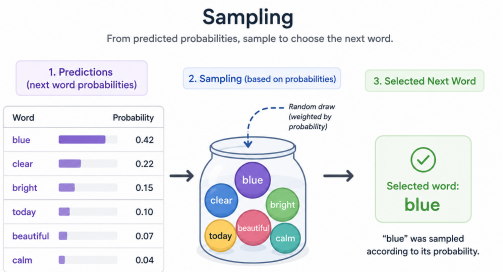
FOUNDATION

What is Sampling & Why It Matters

Every time an LLM generates a token, it's making a probabilistic choice not a deterministic one.

The model doesn't "write" text. It assigns probability scores to every possible next token, then picks one.

How it picks that token is controlled by sampling parameters.



Real Scenario

You're building a legal document summarizer.

Wrong settings on the legal tool = hallucinated clauses.

A colleague uses the same model for a creative story generator.

Wrong settings on the story tool = robotic output.

Same model. Completely different sampling settings. Completely different behavior.

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How Token Probability Works

Before explaining the parameters you need to understand what they operate on.

SCENARIO

Model is mid-sentence: "The capital of France is ___"

"Paris"	94%
"Lyon"	2%
"France"	1.5%
"a"	0.8%

→ Without any sampling model always picks "Paris" (argmax / greedy decoding).

Without sampling the model can pick from the distribution, introducing variety.

WHY GREEDY DECODING FAILS

Always picking the highest probability token leads to repetitive, generic output:
"The best way to learn is to learn by doing and learning from your mistakes..."

PARAMETER 1

Temperature

What it controls: How sharp or flat the probability distribution is before sampling.

↓ **Low temp (0.0 - 0.3)** sharpens the distribution, makes the top token dominate.

↑ **High temp (0.8 - 2.0)** flattens the distribution, gives lower tokens more chance.

Low temperature (0.1)

You ask: "What is the boiling point of water?"
Model is near-certain. "100°C" dominates at 99%+. Low temp keeps it deterministic. Right call.

High temperature (1.2)

You ask: "Write me a tagline for a coffee brand"
High temp explores more of the distribution. More creative variance. Right call.

WHEN TO TUNE

0.0 - 0.3: Factual QA, RAG, legal, medical
0.5 - 0.7: Balanced assistants, chat
0.8 - 1.2: Creative writing, marketing copy


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PARAMETER 2

Top-k Sampling

What it controls: The number of candidate tokens the model considers at each step.

- **Top-k = 1** always pick the highest probability token (greedy).
- **Top-k = 50** only consider the top 50 tokens. Everything else is cut to zero.

"SHE OPENED THE DOOR AND ___"

"walked"	35%	"entered"	25%
"saw"	18%	"found"	12%
"slammed"	6%	"danced"	1.5%

Top-k = 3 only "walked", "entered", "saw" are candidates.
Top-k = 50 includes "danced" and other unlikely picks.

THE PROBLEM WITH TOP-K

k is a fixed number. Sometimes top 50 are all reasonable, sometimes only top 2 make sense. It doesn't adapt to how "certain" the model is.

PARAMETER 3

Top-p (Nucleus)

What it controls: Consider tokens until their cumulative probability reaches p.

THE NUCLEUS

Top-p = 0.9 keep adding tokens from highest to lowest probability until the running total hits 90%. It dynamically sizes the candidate pool.

High Certainty

"The capital of France is ___"
"Paris" is 94%. At top-p = 0.9, only "Paris" is in the nucleus. Correct behavior.

Low Certainty

"She felt ___"
"happy" (18%), "sad" (15%), "nervous" (12%)...
At top-p = 0.9, many tokens are needed to reach 90%. Wide nucleus. More creative picks.

WHY TOP-P BEATS TOP-K

It adapts to the model's confidence automatically. Same parameter, intelligent behavior in both cases.


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How They Interact

These parameters don't work in isolation they stack.

- **Temperature** reshapes the distribution first.
- **Top-k** then cuts it to the top k tokens.
- **Top-p** then cuts further to the nucleus.

BAD CONFIG

```
temp=1.5, top_p=0.95,
top_k=500
```

Too much randomness. Bot adds creative speculation to factual answers.

GOOD CONFIG

```
temp=0.2, top_p=0.5,
top_k=20
```

Tight, deterministic, grounded. Same query = same accurate answer.

Common Mistake: Setting high temp AND high top-p AND high top-k together compounds into maximum randomness. Pick one lever to push high.


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Real-World Recipes

Legal Summarizer

Accurate, consistent, zero hallucination.

`T: 0.1 | P: 0.3 | K: 10`

Support Chatbot

Helpful & natural, but factually grounded.

`T: 0.4 | P: 0.7 | K: 50`

Marketing Copy

Creative, punchy, varied outputs.

`T: 0.9 | P: 0.92 | K: 200`

Code Assistant

Correct & consistent. Creativity is a bug.

`T: 0.1 | P: 0.2 | K: 5`

Creative Story Writer

Maximum variety, unexpected turns, rich language.

`T: 1.2 | P: 0.95 | K: 500`



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Parameter Cheatsheet

Parameter	Low Value	High Value
Temperature Distribution	Focused	Creative
Top-k Fixed count	Few options	Many options
Top-p Mass cutoff	Tight nucleus	Wide nucleus

Use Case	Temp	Top-p	Top-k
RAG / QA	0.1	0.3	10
Support	0.4	0.7	50
Code Gen	0.1	0.2	5
Marketing	0.9	0.92	200
Creative	1.2	0.95	500

★ GOLDEN RULES

- When correctness matters turn down all three.
- When creativity matters temp is your primary lever, top-p secondary.
- Never max out all three simultaneously.


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